

RISKPULSE “ FEATURE WORKFLOW ATLAS

Comprehensive Operational & Research GIS Workflow Reference

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WORKFLOW 1: OPERATIONAL RISKMAP NAVIGATION & ATTRIBUTE QUERY

Module: `Operational RiskMap` | Status: `IMPLEMENTED & SOFTWARE-VERIFIED`

PURPOSE: View live/historical operational disaster risk points and hazard polygons across Mandi, Kinnaur, and Uttarakhand.

INPUTS: Protected production GeoJSON dataset (168 features, Kotropi anchor).

PRECONDITIONS: RiskPulse app launched in Operational Mode.

USER ACTIONS: 1. Open Operational RiskMap. 2. Tap on any risk point or hazard polygon. 3. Toggle basemap style.

PROCESSING: Filters feature properties, calculates simulated risk scores based on susceptibility and live weather factor.

OUTPUTS: Interactive feature attribute card displaying district name, hazard category, susceptibility score, and historical event details.

INTERPRETATION: Provides operational situational awareness for public and emergency managers.

LIMITATIONS: Operational layers are protected from experimental research modifications.

PROVENANCE: Backing feature IDs and GeoJSON property schema.

EXPORT / NEXT STEP: Click [VIEW FULL DETAILS] or open Emergency Hub for response coordination.

WORKFLOW 2: RESEARCH GIS SESSION INITIALIZATION & AOI CAPTURE

Module: `Research GIS Studio` | Status: `IMPLEMENTED & SOFTWARE-VERIFIED`

PURPOSE: Establish a new cartographic research session and capture Study Area / AOI

extent.

****INPUTS**:**

Researcher-selected map extent bounds (southWest, northEast).

****PRECONDITIONS**:**

Open Research GIS Studio screen.

****USER ACTIONS**:**

1. Click [NEW RESEARCH SESSION].
2. Zoom/pan map to target study area.
3. Click [CAPTURE AOI].

****PROCESSING**:**

Creates ResearchSession, sets MapComposition extent, and registers prod-aoi in Product Registry.

****OUTPUTS**:**

Active Research Session with defined CRS (EPSG:4326) and extent.

****INTERPRETATION**:**

Defines spatial domain boundary for all downstream DEM, HydroAI, and RS analyses.

****LIMITATIONS**:**

AOI extent bounds dictate raster cropping and validation grid limits.

****PROVENANCE**:**

Recorded in ResearchWorkspaceProvider state.

****EXPORT / NEXT STEP**:**

Proceed to DEM Acquisition or Remote Sensing ingestion.

WORKFLOW 3: DEM ACQUISITION, STRUCTURAL VALIDATION & READINESS GATING

****Module**:** `DEM Governance` | ****Status**:** `IMPLEMENTED & SOFTWARE-VERIFIED`

****PURPOSE**:**

Acquire DEM raster (GEE GLO-30 / Local GeoTIFF), validate structure (4K.8.12-v1), and evaluate readiness policy (4K.8.14-v1).

****INPUTS**:**

Raw DEM raster (GeoTIFF) or GEE GLO-30 tile.

****PRECONDITIONS**:**

Active Research Session with captured AOI.

****USER ACTIONS**:**

1. Click [ACQUIRE DEM].
2. Select GEE GLO-30 or upload local GeoTIFF.
3. Review validation statistics.
4. If requiresResearcherReview, click [ACKNOWLEDGE & PROCEED].

****PROCESSING**:**

DemValidationService computes footprint coverage %, valid cell %, NoData %.
DemReadinessPolicyService evaluates readiness status.

****OUTPUTS**:**

Governed inputDem raster, DemReadinessAssessment, and
DemReadinessAcknowledgementDialog log.

****INTERPRETATION**:**

Ensures DEM is structurally valid prior to running HydroAI or terrain services.

****LIMITATIONS**:**

Acknowledgement is a workflow decision, NOT objective scientific validation.
Zero elevation fabrication or NoData filling occurs.

****PROVENANCE**:**

validationRuleVersion (4K.8.12-v1) and readinessPolicyVersion (4K.8.14-v1)
recorded in AnalyticalStep.

****EXPORT / NEXT STEP**:**

Proceed to Terrain Analysis or HydroAI pipeline execution.

WORKFLOW 4: HYDROAI 2D HYDRODYNAMIC SIMULATION EXECUTION

****Module**:** `HydroAI Hydrodynamics` | ****Status**:** `SOFTWARE CONTRACTS & ADAPTER
COMPLETE / NATIVE BINARY PENDING`

****PURPOSE**:**

Execute 2D hydrodynamic flood simulation using solver-neutral contracts and
HecRasSolverAdapter.

****INPUTS**:**

Governed inputDem, precipitation forcing (HazardTimeSeries), roughness raster,
and 1D channel geometry.

****PRECONDITIONS**:**

Governed DEM acquired and validated.

****USER ACTIONS**:**

1. Configure HydrodynamicModelDomain.
2. Click [RUN HYDROAI SIMULATION].
3. Monitor execution status.

****PROCESSING**:**

HecRasInputTranslator generates .prj, .g01, .u01, .p01 files.
HecRasProcessController manages execution. HecRasOutputParser parses output
rasters.

****OUTPUTS**:**

HydrodynamicResult containing FloodDepthRaster (Max), VelocityVectorRaster, WSE
raster, and temporal FloodState stack.

****INTERPRETATION**:**

Provides physics-based 2D flood depth and flow velocity fields.

****LIMITATIONS**:**

Simulation completion (SimulationState.completed) indicates software completion ONLY. scientificStatus remains provisionalSoftwareOnly.

****PROVENANCE**:**

AnalyticalStep chains simulationId, solverName, planFile, and DEM readiness metadata.

****EXPORT / NEXT STEP**:**

Proceed to SAR 2D Inundation Validation or Exposure Intersection.

WORKFLOW 5: SENTINEL-1 SAR 2D INUNDATION VALIDATION & CSI EVALUATION

****Module**:** `SAR Validation` | ****Status**:** `SOFTWARE ENGINE COMPLETE / EMPIRICAL EXPERIMENTS PENDING`

****PURPOSE**:**

Perform 2D spatial validation comparing HydroAI flood depth against independent Sentinel-1 SAR observed flood mask.

****INPUTS**:**

Completed HydrodynamicResult, SarInundationRecord, and explicit depthThresholdMeters (e.g. 0.50m).

****PRECONDITIONS**:**

HydrodynamicResult and SAR reference raster share Common Validation Grid.

****USER ACTIONS**:**

1. Open SAR Validation panel.
2. Select SAR dataset and supply wetting threshold (depthThresholdMeters).
3. Click [RUN SAR VALIDATION].

****PROCESSING**:**

SarInundationValidationEngine verifies grid compatibility, derives model extent mask, and computes confusion matrix (TP, FP, FN, TN) and $CSI = TP / (TP + FP + FN)$.

****OUTPUTS**:**

InundationValidationRecord, 4-class categorical map overlay (TP: Green, FP: Orange, FN: Red), and HydroaiValidationPanel metrics.

****INTERPRETATION**:**

Quantifies spatial extent agreement between modeled flood and satellite observation.

****LIMITATIONS**:**

SAR flood extent is an independent reference, NOT absolute ground truth. CSI calculation does NOT constitute automatic scientific validation or operational promotion.

****PROVENANCE**:**

Validation parameters, grid status, and confusion matrix counts recorded in

AnalyticalStep.

****EXPORT / NEXT STEP**:**

Inspect HydroaiValidationPanel metrics and export validation report.

WORKFLOW 6: ENVIRONMENTAL HEALTH & DISEASE SPATIAL INTELLIGENCE ANALYSIS

****Module**:** `Environmental Health` | ****Status**:** `IMPLEMENTED & SOFTWARE-VERIFIED (EH.1-R1)`

****PURPOSE**:**

Analyze spatial geographic associations between aggregated disease incidence rates and environmental exposure variables.

****INPUTS**:**

HealthOutcomeDataset (privacy-safe aggregated district/block cases) and EnvironmentalExposureLayer (e.g. Arsenic Groundwater ppm).

****PRECONDITIONS**:**

Active Research Session with minimum 3 valid observation pairs.

****USER ACTIONS**:**

1. Open Environmental Health panel.
2. Select Health Outcome Dataset and Exposure Layer.
3. Click [RUN SPATIAL ASSOCIATION ANALYSIS].
4. Click [LOAD EXPOSURE OVERLAY TO MAP].

****PROCESSING**:**

EnvironmentalHealthService calculates Pearson correlation r and Student t -statistic p -value over n usable observation pairs.

****OUTPUTS**:**

HealthSpatialAnalysisResult, Exploratory Exposure-Health Overlay raster, and EnvironmentalHealthPanel metrics.

****INTERPRETATION**:**

Identifies spatial geographic associations ONLY.

****LIMITATIONS**:**

Spatial association identified. Further epidemiological investigation is required. Statistical association does NOT establish causation. Zero patient PII exposed.

****PROVENANCE**:**

analysisId, pearsonCorrelationR, sampleCountN, pValue, and non-causality disclaimer recorded in AnalyticalStep.

****EXPORT / NEXT STEP**:**

Export overlay raster or compile epidemiological research briefing.
